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Catherine Chen

Summary of Skills and QualiFications

- Master's candidate in Financial Insurance with hands-on credit risk (EAD/LCF), longevity risk, and Monte Carlo modelling experience
- Skilled in financial analysis, quantitative risk, statistics, and ML, with communication, problem-solving, and multitasking abilities
- Proficient in Python, R, SQL, MATLAB, Google Analytics, Excel/VBA, and Power BI for data analysis and automation

Education

- **Candidate, Master of Financial Insurance (MFI)** Sep. 2026 - Present (Expected completion 2027)
  University of Toronto, Department of Statistical Sciences
  Relevant Courses: Mathematical Finance, Insurance Risk Management, Data Science and Machine Learning
- **Bachelor of Mathematics, Co-op Program (With Distinction)** Sep. 2021 - Jun. 2026
  University of Waterloo, Department of Statistics and Actuarial Science
  Honours Mathematics/Financial Analysis and Risk Management (FARM)

Professional Experience

- **Procurement & Strategic Operations (Co-op), Municipal Property Assessment Corporation (MPAC)**
  Pickering, Canada Apr. 2025 – Aug. 2025
  • Developed an **insurance compliance dashboard** for 120+ suppliers, automating expiry alerts and cutting follow-ups by 75%
  • Co-led a **workflow survey** across 6 business units, using descriptive analytics to identify bottlenecks and improve collaboration
  • Built a **supplier risk dataset** from EDIA and Code of Conduct disclosures, enabling a heatmap for 120+ suppliers
- **Insurance and Enterprise Risk (Co-op), Toronto District School Board (TDSB)**
  North York, Canada May 2024 – Dec. 2024
  • Automated tracking for 200+ vendors with **Excel formulas**, reducing manual work across onboarding, renewals, and approvals
  • Built **Excel VBA** to classify email inquiries and summarize recurring issues, improving response efficiency and transparency
  • Maintained the Excursion website using **HTML/CSS**, resolving technical issues and redesigning layout to improve usability
- **Web Assistant (Co-op), University of Waterloo**
  Waterloo, Canada Sep. 2022 – Dec. 2022 & May 2023 – Aug. 2023
  • Modernized 100+ pages (Work-Learn Institute), including **link validation**, **layout redesign**, and bug fixes using **HTML/CSS**
  • Built responsive pages from wireframes and refined layouts for Centre for Work-Integrated Learning in a 6-person team
  • Developed **Python** scripts to collect web data and **Google Analytics** for engagement analysis, improving web maintenance efficiency

Research & Technical Projects

- **American Put Option Pricing and Early Exercise Analysis, University of Toronto**
  Toronto, Canada Nov. 2026
  • Implemented **CRR binomial pricing in Python (NumPy)** and derived variational **PDE conditions** for American put options
  • Analysed **early exercise boundaries and delta hedging**, testing price convergence across interest-rate and volatility scenarios
  • Used **Monte Carlo simulation** with 10,000 paths to assess discounted PL, exercise timing and volatility misspecification
- **Exposure at Default and Credit Conversion Factor Analysis, University of Toronto**
  Toronto, Canada Oct. 2026
  • Cleaned and integrated six source tables in **Python (pandas)**, building an analytical dataset of 4,733 revolving default episodes
  • Calculated **EAD and CCF** from historical balances, credit limits and pending authorizations using a 180-day observation horizon
  • Checked **historical record timing** and reconciled EAD/CCF calculations, with no duplicate episode IDs or formula discrepancies
- **Mortality Pooling and Longevity Risk in Life Annuity Pricing, University of Toronto**
  Toronto, Canada Jul. 2026 – Sep. 2026
  • Applied **analytical annuity pricing** to compare individual and pooled mortality risk for life annuities at a 4% interest rate
  • Performed 10,000 **Monte Carlo simulations**, with the simulated mean within 0.083% of the analytical expected present value
  • Applied **longevity sensitivity analysis**: a 9% mortality reduction increased benefit liability from $91,576 to $93,523 per policy
- **Machine Learning-Based Evaluation of Small and Micro Business Projects, East China Normal University**
  Shanghai, China Sep. 2023 – Dec. 2023
  • Used **data encoding and manual annotation** to structure expert reviews from 314 articles across six evaluation dimensions
  • Developed an **MLP model in R (RSNNS)** to predict project investment recommendations, achieving 81.8% validation accuracy
  • Applied **data cleaning and outlier filtering**, then split training and validation sets to evaluate model performance